

WILP, BITS Pilani



MATHEMATICAL FOUNDATIONS FOR MACHINE LEARNING · PRACTICE SET

Constrained Optimization

Lagrange multipliers, KKT conditions, and duality — 15 fully worked problems

Frame → solve → dualize → verify · Every algebraic step shown · Attempt each problem before reading its solution

Prepared for the Work Integrated Learning Programmes (WILP), BITS Pilani.

How to use this set. Each problem sits in a **navy** box. Try it on paper first. Then read the solution, which shows every algebraic step — nothing is skipped. The **green** box states the final answer. **Blue** boxes explain the idea behind a step, **red** boxes flag the classic mistakes, and **purple** boxes hold the one lesson each problem teaches. The problems build on each other: Part A uses equality constraints only, Part B brings in inequalities and KKT, Part C converts primals to duals and solves them, and Part D uses complementary slackness to test candidate answers — ending with a tiny support vector machine that uses everything at once.

1 Read this first: the toolkit

Four recipes solve every problem in this set. Everything else is careful bookkeeping.

Standard form. Before doing anything, rewrite the problem as a **minimization in standard form**:

$$\min_x f(x) \quad \text{subject to} \quad g_i(x) \leq 0 \quad (i = 1, \dots, m), \quad h_j(x) = 0 \quad (j = 1, \dots, p).$$

The conversion rules are mechanical: a maximization $\max f$ becomes $\min(-f)$; a constraint $a \geq b$ becomes $b - a \leq 0$; an equality $h = c$ becomes $h - c = 0$.

The intuition

Everything in this topic answers one question: *at the best feasible point, how does the objective's gradient relate to the constraints' gradients?* Lagrange multipliers are the bookkeeping device that records that relationship — one multiplier per constraint, measuring how hard that constraint “pushes”.

Recipe 1 — Lagrange (equality constraints only). To optimize $f(x)$ subject to $h(x) = 0$: at a constrained optimum the two gradients are parallel,

$$\nabla f(x) = \lambda \nabla h(x), \quad h(x) = 0.$$

Solve these equations together. With several equality constraints $h_1 = \dots = h_p = 0$, the condition becomes $\nabla f = \lambda_1 \nabla h_1 + \dots + \lambda_p \nabla h_p$. The sign of λ is unrestricted here — an equality constraint can push either way.

Recipe 2 — KKT (the full set of conditions). For the standard form above, build the **Lagrangian**

$$L(x, \mu, \lambda) = f(x) + \sum_{i=1}^m \mu_i g_i(x) + \sum_{j=1}^p \lambda_j h_j(x).$$

A point x^* with multipliers (μ^*, λ^*) satisfies the **KKT conditions** when all four hold:

K1. Stationarity: $\nabla_x L(x^*, \mu^*, \lambda^*) = 0$.

K2. Primal feasibility: $g_i(x^*) \leq 0$ for all i , and $h_j(x^*) = 0$ for all j .

K3. Dual feasibility: $\mu_i^* \geq 0$ for all i .

K4. Complementary slackness: $\mu_i^* g_i(x^*) = 0$ for every i .

For a *convex* problem (convex f and g_i , affine h_j), any point satisfying K1–K4 is a *global* minimizer — KKT is sufficient. Under a constraint qualification such as Slater’s condition, KKT is also necessary.

Recipe 3 — the complementary-slackness case method. Condition K4 says every inequality constraint is in one of two states: **inactive** ($\mu_i = 0$, the constraint exerts no force) or **active** ($g_i(x^*) = 0$, the point sits on the boundary). With m inequalities there are at most 2^m active/inactive patterns. Pick a pattern, solve K1 together with the assumed active equations, then *check* what the pattern did not enforce: feasibility of the constraints assumed inactive, and $\mu_i \geq 0$ for the constraints assumed active. Exactly the patterns that pass every check survive.

Recipe 4 — Lagrangian duality. From the same Lagrangian, define the **dual function** by minimizing over x with the multipliers held fixed:

$$q(\mu, \lambda) = \inf_x L(x, \mu, \lambda).$$

This inner minimization is *unconstrained* — that is the whole point. The **dual problem** is $\max_{\mu \geq 0, \lambda} q(\mu, \lambda)$, and it is always concave. Writing p^* for the primal optimum and d^* for the dual optimum: **weak duality** $d^* \leq p^*$ holds for *every* problem; **strong duality** $d^* = p^*$ holds for convex problems satisfying **Slater’s condition** (some strictly feasible point exists, $g_i(\tilde{x}) < 0$ for all i). Under strong duality, the primal solution is recovered from the inner minimization evaluated at the optimal multipliers.

⚠ Watch out

KKT as written above is for **minimization** with $\leq$ constraints. Applying it to a maximization, or to a $\geq$ constraint, without converting first is the single most common source of sign errors in this topic. Convert to standard form *before* writing the Lagrangian — always. Problems 7, 9 and 10 drill exactly this.

The 15 problems at a glance

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2 Part A — Framing problems and the Lagrange condition

Turn a story into “optimize f subject to $h = 0$ ”, then solve $\nabla f = \lambda \nabla h$ together with the constraint.

Problem 1 · Framing a word problem, then solving it

A farmer has 20 metres of fencing and wants to enclose a rectangular plot of the largest possible area.

- Frame this as a constrained optimization problem: name the decision variables, the objective, and the constraint.
- Solve it with a Lagrange multiplier.

Skill trained: translating words into $\max f$ subject to $h = 0$, then applying Recipe 1.

Solution

Step 1. Name the variables. Let $x > 0$ be the width and $y > 0$ the height of the plot, both in metres.

Step 2. Write the objective. The area is $A(x, y) = xy$, and we want it as large as possible: $\max_{x,y} xy$.

Step 3. Write the constraint. The perimeter uses all the fencing: $2x + 2y = 20$. Dividing by 2 gives the cleaner equivalent $h(x, y) = x + y - 10 = 0$. The framed problem is

$$\max_{x,y} xy \quad \text{subject to} \quad x + y = 10.$$

Step 4. Write the Lagrange condition. $\nabla f = \lambda \nabla h$ with $f = xy$ and $h = x + y - 10$. The gradients are $\nabla f = (y, x)$ and $\nabla h = (1, 1)$, so

$$\begin{aligned} y &= \lambda \cdot 1 = \lambda, \\ x &= \lambda \cdot 1 = \lambda. \end{aligned}$$

Step 5. Solve the system. Both equations say the same thing: $x = y = \lambda$. Substitute into the constraint:

$$x + y = 10 \implies \lambda + \lambda = 10 \implies \lambda = 5.$$

So $x = 5$ and $y = 5$: the optimal plot is a square.

Step 6. Confirm it is a maximum. On the constraint line, $y = 10 - x$ and the area becomes $A(x) = x(10 - x) = 10x - x^2$, a downward parabola. Its only critical point is the top, and at the endpoints ($x \rightarrow 0$ or $x \rightarrow 10$) the area collapses to 0. So $x = y = 5$ is the maximum, with $A = 5 \times 5 = 25$.

Final answer

Decision variables x, y ; objective $\max xy$; constraint $x + y = 10$. Optimum: $x^* = y^* = 5$ m (a square), maximum area $A^* = 25 \text{ m}^2$, multiplier $\lambda^* = 5$.

✓ Key takeaway

Framing is three sentences: *what can I choose* (variables), *what do I want* (objective), *what limits me* (constraints). Only after those three lines does any calculus start.

🔖 Problem 2 · Geometry of the Lagrange condition

Find the point on the line $x + y = 4$ that is closest to the origin, by minimizing $f(x, y) = x^2 + y^2$ subject to $x + y = 4$. Then explain, in one sentence, what $\nabla f = \lambda \nabla h$ means geometrically at your answer.

Skill trained: solving Recipe 1 cleanly, and reading the tangency picture behind it.

Solution

Step 1. Set up the condition. Here $f = x^2 + y^2$ and $h = x + y - 4$, so $\nabla f = (2x, 2y)$ and $\nabla h = (1, 1)$. The Lagrange condition $\nabla f = \lambda \nabla h$ gives

$$\begin{aligned} 2x &= \lambda, \\ 2y &= \lambda. \end{aligned}$$

Step 2. Eliminate λ . Subtracting the two equations: $2x - 2y = 0$, so $x = y$.

Step 3. Use the constraint. $x + y = 4$ with $x = y$ gives $2x = 4$, so $x = 2$ and $y = 2$. Then $\lambda = 2x = 4$.

Step 4. Compute the minimum value. $f(2, 2) = 2^2 + 2^2 = 8$. As a check, the distance from the origin to the line $x + y = 4$ is $|-4|/\sqrt{1^2 + 1^2} = 4/\sqrt{2} = 2\sqrt{2}$, and $(2\sqrt{2})^2 = 8$. The two computations agree.

Step 5. Read the geometry. At $(2, 2)$ the level curve of f (the circle $x^2 + y^2 = 8$) just touches the line: the circle is *tangent* to the constraint. Tangency is exactly the statement that the two gradients point along the same direction — $\nabla f = (4, 4) = 4 \cdot (1, 1) = \lambda \nabla h$.

✓ Final answer

Closest point $(x^*, y^*) = (2, 2)$, with $\lambda^* = 4$ and minimum value $f^* = 8$ (distance $2\sqrt{2}$). Geometrically: at the optimum, the level curve of f is tangent to the constraint, so ∇f and ∇h are parallel.

🔖 The intuition

If ∇f had any component *along* the constraint line, you could slide along the line in that direction and improve f — so at the optimum, no such component can remain. That forces ∇f to be perpendicular to the line, i.e. parallel to ∇h . This one picture is the entire logic of Lagrange multipliers.

Tangency: the smallest circle that still touches the line

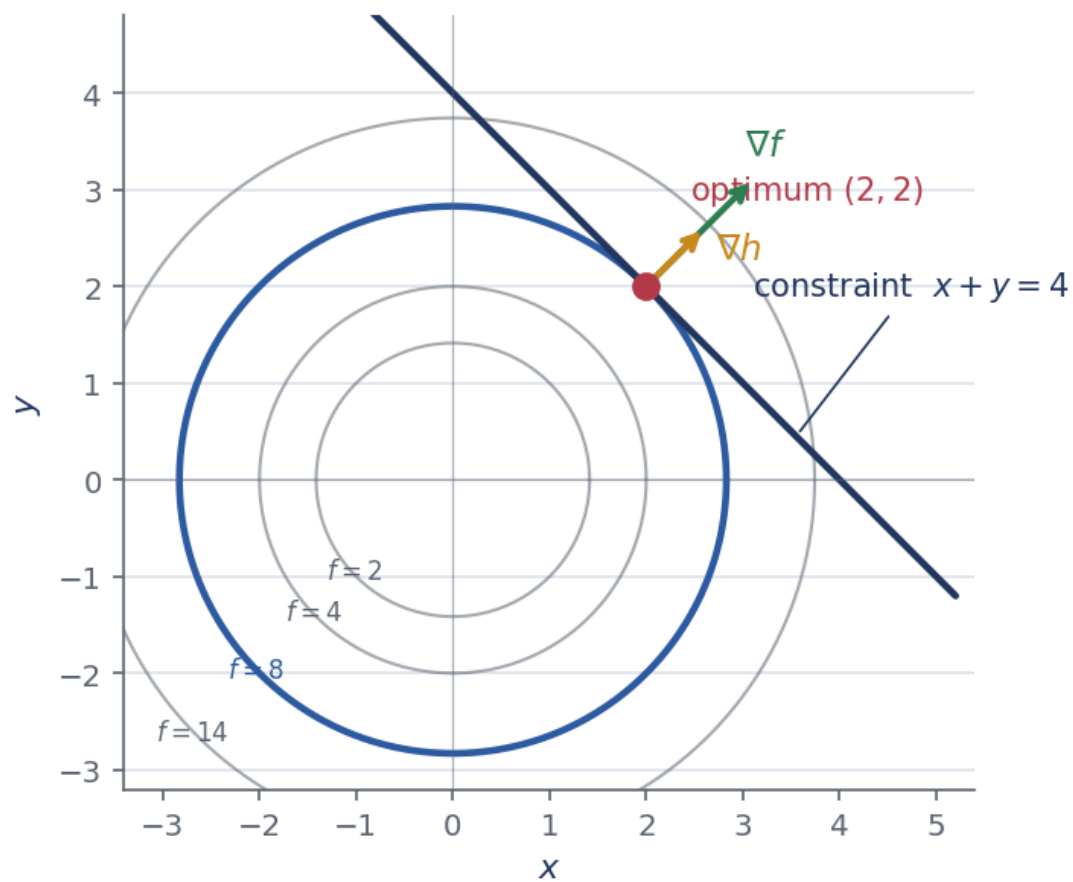


Figure 1: How to read it: the smallest circle that still touches the line is the level set $f = 8$; at the touching point $(2, 2)$ the gradient of f and the gradient of the constraint are parallel.

Problem 3 · The multiplier is a shadow price

A workshop produces $Q(x, y) = xy$ units using x machine-hours (cost Rs. 2 per hour) and y labour-hours (cost Rs. 1 per hour). The budget is Rs. 8.

- (a) Frame and solve: how should the budget be split to maximize output?
 (b) Interpret the multiplier λ . Then re-solve with a budget of Rs. 9 and check the interpretation numerically.

Skill trained: framing with costs, and reading λ as “extra objective per extra unit of budget”.

Solution

Step 1. Frame. Maximize $Q = xy$ subject to the budget constraint $2x + y = 8$, with $x, y > 0$.

Step 2. Lagrange condition. $\nabla Q = (y, x)$ and, for $h = 2x + y - 8$, $\nabla h = (2, 1)$. So $\nabla Q = \lambda \nabla h$ reads

$$\begin{aligned} y &= 2\lambda, \\ x &= \lambda. \end{aligned}$$

Step 3. Eliminate λ . From the second equation $\lambda = x$; substitute into the first: $y = 2x$. In words: at the optimum, spend the same amount on each input ($2x$ rupees on machines equals $y \cdot 1 = 2x$ rupees on labour).

Step 4. Use the budget. $2x + y = 8$ with $y = 2x$:

$$2x + 2x = 8 \implies 4x = 8 \implies x = 2, \quad y = 4.$$

Output $Q = 2 \times 4 = 8$, and $\lambda = x = 2$.

Step 5. Interpret λ . The multiplier is the **shadow price** of the budget: one extra rupee of budget buys approximately $\lambda = 2$ extra units of output.

Step 6. Check with budget 9. Redo Step 4 with $2x + y = 9$ and $y = 2x$: $4x = 9$, so $x = 9/4$, $y = 9/2$, and

$$Q = \frac{9}{4} \cdot \frac{9}{2} = \frac{81}{8} = 10.125.$$

The gain is $10.125 - 8 = 2.125$, close to $\lambda = 2$ as predicted. (In general $x = b/4$, $y = b/2$, so $Q(b) = b^2/8$ and $dQ/db = b/4 = 2$ at $b = 8$ — exactly λ . The prediction is exact in the derivative and approximate over a whole-rupee jump.)

✓ Final answer

$x^* = 2$ machine-hours, $y^* = 4$ labour-hours, $Q^* = 8$ units, $\lambda^* = 2$. The multiplier is the marginal value of budget: $dQ^*/db = \lambda$.

✱ Wait — really?

The multiplier you computed as a mere “helper variable” turns out to be a forecast: it predicts how much the *optimal* objective improves per unit of constraint relaxation — before you re-solve anything. This is why economists call it a price, and why in ML the multipliers on data constraints tell you which data points matter (Problem 15 makes this literal).

🔖 Problem 4 · Two equality constraints at once

Minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to

$$h_1 : x + y + z = 3 \quad \text{and} \quad h_2 : x - y = 0.$$

Skill trained: running Recipe 1 with two multipliers: $\nabla f = \lambda_1 \nabla h_1 + \lambda_2 \nabla h_2$.

Solution

Step 1. Write the condition with two multipliers. With $\nabla f = (2x, 2y, 2z)$, $\nabla h_1 = (1, 1, 1)$ and $\nabla h_2 = (1, -1, 0)$, the condition $\nabla f = \lambda_1 \nabla h_1 + \lambda_2 \nabla h_2$ reads

$$2x = \lambda_1 + \lambda_2,$$

$$2y = \lambda_1 - \lambda_2,$$

$$2z = \lambda_1.$$

Step 2. Use the second constraint. h_2 says $x = y$. Subtract the second equation from the first:

$$2x - 2y = (\lambda_1 + \lambda_2) - (\lambda_1 - \lambda_2) = 2\lambda_2.$$

The left side is 0 (because $x = y$), so $\lambda_2 = 0$.

Step 3. Simplify. With $\lambda_2 = 0$ the three equations collapse to $2x = 2y = 2z = \lambda_1$, i.e. $x = y = z = \lambda_1/2$.

Step 4. Use the first constraint. $x + y + z = 3$ with all three equal gives $3x = 3$, so $x = y = z = 1$ and $\lambda_1 = 2x = 2$.

Step 5. Value. $f(1, 1, 1) = 1 + 1 + 1 = 3$. Since f is a sum of squares (strictly convex) and both constraints are affine, this critical point is the unique global minimum.

✓ Final answer

$(x^*, y^*, z^*) = (1, 1, 1)$, $f^* = 3$, with multipliers $\lambda_1^* = 2$ and $\lambda_2^* = 0$.

🔖 The intuition

$\lambda_2 = 0$ is not an accident. Minimizing distance-to-origin on the plane $x + y + z = 3$ alone already lands on the symmetric point $(1, 1, 1)$, which happens to satisfy $x = y$ for free. A constraint that the solution satisfies anyway exerts no force — its multiplier is zero. This

is complementary slackness making an early, informal appearance.

 Problem 5 · Quadratic forms on the unit circle: hello, eigenvalues

Let $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$. Find the maximum and minimum of the quadratic form

$$f(x, y) = \begin{pmatrix} x & y \end{pmatrix} A \begin{pmatrix} x \\ y \end{pmatrix} = 3x^2 + 2xy + 3y^2$$

subject to $x^2 + y^2 = 1$. What familiar linear-algebra objects do the solutions turn out to be?

Skill trained: seeing the Lagrange condition become the eigenvalue equation $A\mathbf{v} = \lambda\mathbf{v}$.

Solution

Step 1. Gradients. $f = 3x^2 + 2xy + 3y^2$ gives $\nabla f = (6x + 2y, 2x + 6y)$. The constraint $h = x^2 + y^2 - 1$ gives $\nabla h = (2x, 2y)$.

Step 2. Lagrange condition. $\nabla f = \lambda \nabla h$:

$$6x + 2y = 2\lambda x,$$

$$2x + 6y = 2\lambda y.$$

Divide both equations by 2:

$$3x + y = \lambda x,$$

$$x + 3y = \lambda y.$$

Step 3. Recognize the structure. In matrix form this is exactly

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{— the eigenvalue equation.}$$

The candidates are the *unit eigenvectors* of A , and at any such candidate the objective equals $f = \mathbf{v}^\top A \mathbf{v} = \mathbf{v}^\top (\lambda \mathbf{v}) = \lambda \|\mathbf{v}\|^2 = \lambda$. So the constrained max/min of f are the largest/smallest eigenvalues.

Step 4. Find the eigenvalues.

$$\begin{aligned} \det(A - \lambda I) &= (3 - \lambda)^2 - 1 \\ &= \lambda^2 - 6\lambda + 9 - 1 \\ &= \lambda^2 - 6\lambda + 8 = (\lambda - 2)(\lambda - 4) = 0, \end{aligned}$$

so $\lambda = 4$ or $\lambda = 2$.

Step 5. Find the unit eigenvectors. For $\lambda = 4$: $(A - 4I)\mathbf{v} = 0$ gives $-x + y = 0$, so $y = x$ and the unit vector is $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)$. For $\lambda = 2$: $x + y = 0$, so $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1)$.

Step 6. Conclude. Maximum $f = 4$ at $\pm \frac{1}{\sqrt{2}}(1, 1)$; minimum $f = 2$ at $\pm \frac{1}{\sqrt{2}}(1, -1)$. Sanity check at $\frac{1}{\sqrt{2}}(1, 1)$: $3 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} + 3 \cdot \frac{1}{2} = \frac{3+2+3}{2} = 4$. ✓

✓ **Final answer**

Maximum 4 at $\pm(1, 1)/\sqrt{2}$; minimum 2 at $\pm(1, -1)/\sqrt{2}$. The optimizers are the unit eigenvectors of A and the optimal values are its eigenvalues.

✓ **Key takeaway**

“Maximize a quadratic form on the unit sphere” is the eigenvalue problem — this is precisely how PCA finds its first principal component: maximize the variance $\mathbf{v}^\top \Sigma \mathbf{v}$ subject to $\|\mathbf{v}\| = 1$, and the answer is the top eigenvector of the covariance matrix Σ .

3 Part B — Inequality constraints and the KKT conditions

Convert to standard form, write the Lagrangian, then let complementary slackness split the search into clean cases.

Problem 6 · First contact with KKT: active vs. inactive

Solve the two one-dimensional problems

(a) $\min_x (x - 3)^2$ subject to $x \leq 1$;

(b) $\min_x (x - 3)^2$ subject to $x \leq 5$.

For each, write all four KKT conditions explicitly and identify whether the constraint is active or inactive at the optimum.

Skill trained: writing K1–K4 in full, and feeling the difference between a constraint that binds and one that does not.

Solution

Part (a).

Step 1. Standard form. $f(x) = (x - 3)^2$ and $g(x) = x - 1 \leq 0$. Lagrangian: $L(x, \mu) = (x - 3)^2 + \mu(x - 1)$.

Step 2. Write the four KKT conditions.

$$\begin{aligned} \text{K1 (stationarity): } & 2(x - 3) + \mu = 0 \\ \text{K2 (primal feasibility): } & x - 1 \leq 0 \\ \text{K3 (dual feasibility): } & \mu \geq 0 \\ \text{K4 (compl. slackness): } & \mu(x - 1) = 0 \end{aligned}$$

Step 3. Case $\mu = 0$ (constraint inactive). K1 becomes $2(x - 3) = 0$, so $x = 3$. Check K2: $3 - 1 = 2 \not\leq 0$. Infeasible — discard this case.

Step 4. Case $x = 1$ (constraint active). K1 gives

$$2(1 - 3) + \mu = 0 \implies -4 + \mu = 0 \implies \mu = 4.$$

Check K3: $\mu = 4 \geq 0$. ✓ K2 holds ($1 - 1 = 0$), and K4 holds because $g = 0$. All four conditions pass.

Step 5. Conclude (a). $x^* = 1$, $\mu^* = 4$, $f^* = (1 - 3)^2 = 4$. The constraint is *active*: the unconstrained minimizer $x = 3$ lies outside the feasible set, so the solution is pushed onto the boundary and the multiplier records the force of that push.

Part (b).

Step 1. Standard form. Now $g(x) = x - 5 \leq 0$ and $L(x, \mu) = (x - 3)^2 + \mu(x - 5)$. The four conditions are the same four lines with $x - 5$ in place of $x - 1$.

Step 2. **Case $\mu = 0$.** K1: $2(x - 3) = 0$, so $x = 3$. Check K2: $3 - 5 = -2 \leq 0$. ✓ Feasible! K3 holds ($\mu = 0 \geq 0$) and K4 holds ($\mu = 0$). All four pass — no need to try the other case, since the problem is convex and any KKT point is the global minimum.

Step 3. **Conclude (b).** $x^* = 3$, $\mu^* = 0$, $f^* = 0$. The constraint is *inactive*: the unconstrained minimizer already satisfies it, so it exerts no force.

✓ **Final answer**

(a) $x^* = 1$, $\mu^* = 4$, $f^* = 4$ — constraint active. (b) $x^* = 3$, $\mu^* = 0$, $f^* = 0$ — constraint inactive.

🧠 **The intuition**

Think of μ as the *force* the wall exerts on a ball rolling downhill inside the feasible region. In (a) the ball wants to reach $x = 3$ but the wall at $x = 1$ stops it, pressing back with force $\mu = 4$. In (b) the ball settles at $x = 3$ without ever touching the wall at $x = 5$, so the wall pushes with force 0. Complementary slackness is just: *a wall you are not touching cannot push you*.

🧠 **Problem 7 · KKT in 2-D with a $\geq$ constraint**

Minimize $f(x, y) = x^2 + y^2$ subject to $x + y \geq 2$.

Skill trained: converting a $\geq$ constraint to standard form before touching the Lagrangian, then running the two-case analysis.

Solution

Step 1. **Standard form.** Rewrite $x + y \geq 2$ as $g(x, y) = 2 - x - y \leq 0$. Lagrangian:

$$L(x, y, \mu) = x^2 + y^2 + \mu(2 - x - y).$$

Step 2. **Stationarity (K1).**

$$\begin{aligned}\frac{\partial L}{\partial x} = 2x - \mu = 0 &\implies x = \frac{\mu}{2}, \\ \frac{\partial L}{\partial y} = 2y - \mu = 0 &\implies y = \frac{\mu}{2}.\end{aligned}$$

So $x = y$ at any KKT point.

Step 3. **Case $\mu = 0$ (inactive).** Then $x = y = 0$. Check K2: $g(0, 0) = 2 - 0 - 0 = 2 \not\leq 0$. Infeasible — discard. (The origin, the unconstrained minimizer, lies on the wrong side of the line.)

Step 4. **Case active:** $x + y = 2$. With $x = y$ this gives $2x = 2$, so $x = y = 1$. Recover the multiplier from Step 2: $\mu = 2x = 2 \geq 0$. ✓ All four conditions hold.

Step 5. **Value and check.** $f(1, 1) = 1 + 1 = 2$. Geometrically this is the projection of the origin onto the half-plane $x + y \geq 2$: the closest boundary point is the foot of the perpendicular, which by symmetry is $(1, 1)$.

✓ Final answer

$$(x^*, y^*) = (1, 1), \mu^* = 2, f^* = 2.$$

✗ Watch out

If you skip the conversion and write $L = x^2 + y^2 + \mu(x + y - 2)$ with $\mu \geq 0$ for the constraint $x + y \geq 2$, stationarity gives $x = y = -\mu/2$, which forces $x + y = -\mu \leq 0 < 2$: every candidate is infeasible and the method “mysteriously fails”. The method is fine — the sign was wrong. Standard form first, always.

📖 Problem 8 · Two inequalities: the full four-case enumeration

Minimize $f(x, y) = (x - 2)^2 + y^2$ subject to

$$x + y \leq 1 \quad \text{and} \quad y \geq 0.$$

Enumerate all four active/inactive patterns of the two constraints and solve the problem completely.

Skill trained: Recipe 3 in full: propose a pattern, solve, check what the pattern did not enforce.

Solution

Step 1. Standard form. $g_1(x, y) = x + y - 1 \leq 0$ and $g_2(x, y) = -y \leq 0$. Lagrangian:

$$L = (x - 2)^2 + y^2 + \mu_1(x + y - 1) + \mu_2(-y).$$

Step 2. Stationarity (K1).

$$\begin{aligned} \frac{\partial L}{\partial x} &= 2(x - 2) + \mu_1 = 0 \implies \mu_1 = 2(2 - x), \\ \frac{\partial L}{\partial y} &= 2y + \mu_1 - \mu_2 = 0. \end{aligned}$$

Step 3. Case 1: $\mu_1 = 0, \mu_2 = 0$ (both inactive). From K1: $2(x - 2) = 0$ gives $x = 2$, and $2y = 0$ gives $y = 0$. Check $g_1: 2 + 0 - 1 = 1 \not\leq 0$. Infeasible — discard.

Step 4. Case 2: g_1 active, $\mu_2 = 0$. Constraint: $x + y = 1$. From K1, $\mu_1 = 2(2 - x) = 4 - 2x$ and $2y + \mu_1 = 0$, so

$$y = -\frac{\mu_1}{2} = -\frac{4 - 2x}{2} = x - 2.$$

Substitute into $x + y = 1$:

$$x + (x - 2) = 1 \implies 2x = 3 \implies x = \frac{3}{2}, \quad y = -\frac{1}{2}.$$

Check $g_2: -y = \frac{1}{2} \not\leq 0$. Infeasible (y went negative) — discard.

Step 5. Case 3: $\mu_1 = 0, g_2$ active. Constraint: $y = 0$. From K1 with $\mu_1 = 0$: $2(x - 2) = 0$ gives $x = 2$. Check $g_1: 2 + 0 - 1 = 1 \not\leq 0$. Infeasible — discard.

Step 6. Case 4: both active. $x + y = 1$ and $y = 0$ give $x = 1, y = 0$. Recover the multipliers:

$$\begin{aligned}\mu_1 &= 2(2 - 1) = 2 \geq 0, \checkmark \\ \mu_2 &= 2y + \mu_1 = 0 + 2 = 2 \geq 0. \checkmark\end{aligned}$$

K2 holds ($g_1 = 0, g_2 = 0$), K4 holds (both constraints are tight). All four conditions pass.

Step 7. Conclude. The problem is convex (a convex quadratic over an intersection of half-planes), so the surviving KKT point is the global minimum: $(x^*, y^*) = (1, 0)$ with $f^* = (1 - 2)^2 + 0^2 = 1$.

Two walls, one corner: where the smallest circle meets the wedge

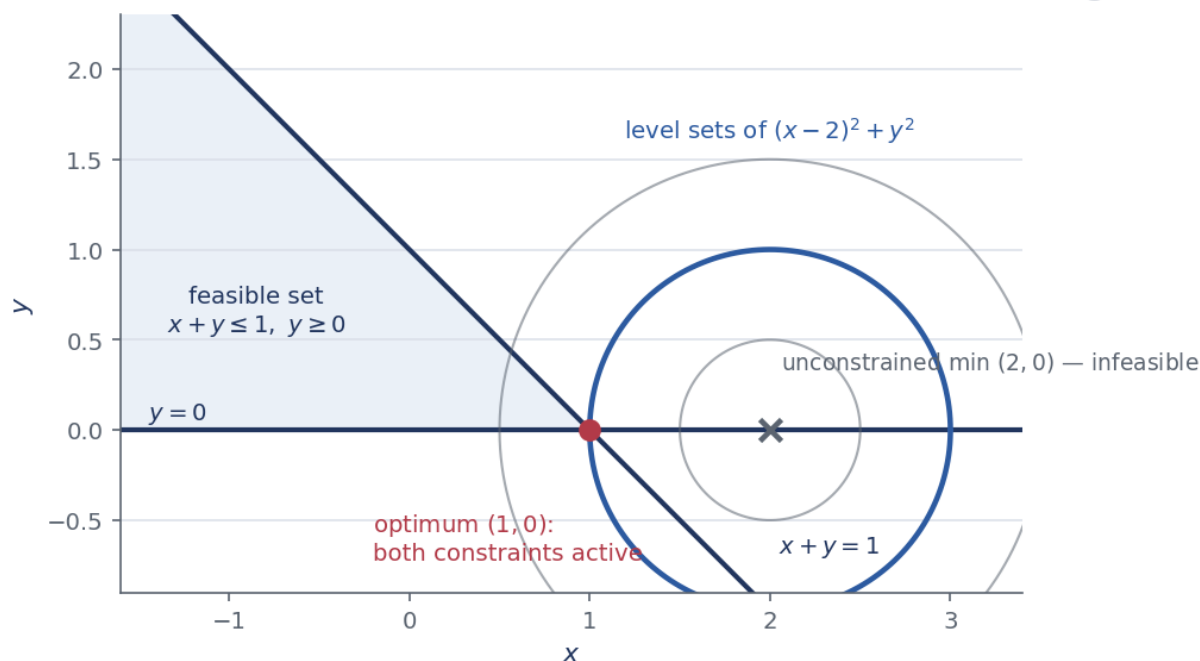


Figure 2: How to read it: the shaded wedge is the feasible set; the circles are level sets of f centred at the unconstrained minimizer $(2, 0)$, which lies outside. The smallest circle that reaches the wedge touches it at the corner $(1, 0)$ — where *both* constraints are active, matching Case 4.

✓ Final answer

Only the both-active pattern survives: $(x^*, y^*) = (1, 0)$, $\mu_1^* = \mu_2^* = 2$, $f^* = 1$.

✓ Key takeaway

The case method is not guesswork — it is exhaustive. Each of the 2^m patterns either *solves and verifies*, or *contradicts itself* (infeasible point, or a negative multiplier). Discarding a case is progress, not failure.

Problem 9 · A maximization handled by KKT

Maximize $f(x) = 4x - x^2$ subject to $x \leq 1$. Solve by first converting to a standard-form minimization, and report the multiplier.

Skill trained: the max-to-min conversion, done without sign accidents.

Solution

Step 1. Convert. $\max (4x - x^2)$ is the same problem as

$$\min_x \tilde{f}(x) = x^2 - 4x \quad \text{subject to} \quad g(x) = x - 1 \leq 0,$$

and the optimal values satisfy $\max f = -\min \tilde{f}$.

Step 2. Lagrangian and K1. $L(x, \mu) = x^2 - 4x + \mu(x - 1)$, so

$$\frac{\partial L}{\partial x} = 2x - 4 + \mu = 0.$$

Step 3. Case $\mu = 0$. $2x - 4 = 0$ gives $x = 2$; check K2: $2 - 1 = 1 \not\leq 0$. Infeasible — discard. (Unconstrained, the parabola peaks at $x = 2$, outside the feasible set.)

Step 4. Case $x = 1$ (active). K1 gives $2(1) - 4 + \mu = 0$, so $\mu = 2 \geq 0$. ✓ All four KKT conditions hold.

Step 5. Report in the original problem. $x^* = 1$ and $\max f = 4(1) - 1^2 = 3$. (Equivalently $\min \tilde{f} = 1 - 4 = -3$ and $\max f = -(-3) = 3$.)

✓ Final answer

$x^* = 1, \mu^* = 2$, maximum value $f^* = 3$.

✗ Watch out

Two equivalent-looking habits, one of them dangerous. Safe: convert to $\min(-f)$ and use the standard machinery untouched. Dangerous: keep the max and “flip the sign of μ in your head”. The second habit works until a problem has three constraints and a $\geq$ among them — then the mental sign-flips collide. This set always converts first.

Problem 10 · KKT on a disk: stationarity can force activity

Maximize $f(x, y) = x$ subject to $x^2 + y^2 \leq 1$.

Skill trained: using the stationarity equation itself to rule out the inactive case — a different discard mechanism than infeasibility.

Solution

Step 1. Standard form. Minimize $\tilde{f}(x, y) = -x$ subject to $g(x, y) = x^2 + y^2 - 1 \leq 0$. Lagrangian:

$$L = -x + \mu (x^2 + y^2 - 1).$$

Step 2. Stationarity (K1).

$$\begin{aligned}\frac{\partial L}{\partial x} &= -1 + 2\mu x = 0, \\ \frac{\partial L}{\partial y} &= 2\mu y = 0.\end{aligned}$$

Step 3. Case $\mu = 0$ dies inside K1. The first equation becomes $-1 = 0$ — a contradiction. So there is *no* KKT point with $\mu = 0$: this time the inactive case is killed by stationarity itself, not by infeasibility. The constraint *must* be active.

Step 4. Active case. $\mu > 0$, so the second equation forces $y = 0$, and K4 (with $\mu > 0$) forces $x^2 + y^2 = 1$, giving $x = \pm 1$. The first equation says $x = \frac{1}{2\mu}$, which is positive (since $\mu > 0$), so $x = +1$ and then $\mu = \frac{1}{2x} = \frac{1}{2} \geq 0$. ✓

Step 5. Conclude. $(x^*, y^*) = (1, 0)$ and the maximum of the original objective is $f^* = 1$. Geometrically: pushing x as far right as the disk allows lands on the rightmost boundary point.

✓ **Final answer**

$$(x^*, y^*) = (1, 0), \mu^* = \frac{1}{2}, \text{ maximum } f^* = 1.$$

👉 **The intuition**

A linear objective has a constant, nonzero gradient — it never “settles” in the interior of a region. So for linear objectives the optimum always lies on the boundary, and KKT knows it: the inactive case self-destructs at stationarity. This is also why linear programs are solved at vertices.

4 Part C — From primal to dual, and back

Minimize the Lagrangian over x to get the dual function $q(\mu)$, maximize q over $\mu \geq 0$, then recover the primal solution.

Problem 11 · Primal to dual, end to end, in one variable

Consider the primal problem

$$\min_x x^2 \quad \text{subject to} \quad x \geq 2.$$

- (a) Write the Lagrangian and derive the dual function $q(\mu)$ in closed form.
- (b) State and solve the dual problem.
- (c) Verify strong duality by solving the primal directly, and recover x^* from the dual solution.

Skill trained: Recipe 4 on the smallest possible example: every move visible.

Solution

Step 1. Standard form and Lagrangian. The constraint $x \geq 2$ becomes $g(x) = 2 - x \leq 0$, so

$$L(x, \mu) = x^2 + \mu(2 - x) = x^2 - \mu x + 2\mu.$$

Step 2. Dual function = unconstrained minimum over x . For fixed μ , L is a convex parabola in x . Set its x -derivative to zero:

$$\frac{\partial L}{\partial x} = 2x - \mu = 0 \implies x = \frac{\mu}{2}.$$

Substitute back:

$$\begin{aligned} q(\mu) &= L\left(\frac{\mu}{2}, \mu\right) = \left(\frac{\mu}{2}\right)^2 - \mu \cdot \frac{\mu}{2} + 2\mu \\ &= \frac{\mu^2}{4} - \frac{\mu^2}{2} + 2\mu = -\frac{\mu^2}{4} + 2\mu. \end{aligned}$$

Note q is a *concave* parabola in μ — dual functions always are.

Step 3. Solve the dual. The dual problem is $\max_{\mu \geq 0} q(\mu) = -\frac{\mu^2}{4} + 2\mu$. Set the derivative to zero:

$$q'(\mu) = -\frac{\mu}{2} + 2 = 0 \implies \mu^* = 4 \quad (\geq 0, \text{ so the constraint } \mu \geq 0 \text{ is satisfied}).$$

$$\text{Dual optimum: } d^* = q(4) = -\frac{16}{4} + 8 = -4 + 8 = 4.$$

Step 4. Solve the primal directly. x^2 decreases toward $x = 0$, so on the feasible set $x \geq 2$ the minimum sits at the boundary: $x^* = 2$, $p^* = 4$.

Step 5. Compare and recover. $d^* = 4 = p^*$: zero duality gap (strong duality — the problem is convex and $x = 3$ is strictly feasible, so Slater holds). The inner minimizer at μ^* recovers the primal solution: $x^* = \mu^*/2 = 4/2 = 2$. ✓

✓ Final answer

$q(\mu) = -\mu^2/4 + 2\mu$; dual solution $\mu^* = 4$ with $d^* = 4$; primal solution $x^* = 2$ with $p^* = 4$. Strong duality holds, and $x^* = \mu^*/2$.

🔍 The intuition

The dual is a bargaining game. The dual variable μ prices violations of the constraint; for each price, the primal side responds with the cheapest x (the inner minimization). A too-low price lets x drift toward 0 and ignore the constraint; a too-high price over-corrects. The dual problem searches for the price at which the primal side, acting selfishly, *voluntarily* lands on the true constrained optimum. Here that price is $\mu^* = 4$.

🔍 Problem 12 · Primal to dual in two variables, plus Slater and recovery

Consider

$$\min_{x,y} x^2 + y^2 \quad \text{subject to} \quad x + y \geq 4.$$

- (a) Derive the dual function $q(\mu)$ and solve the dual problem.
- (b) Verify Slater's condition and confirm $p^* = d^*$.
- (c) Recover (x^*, y^*) from μ^* and verify all four KKT conditions at the primal–dual pair.

Skill trained: the full round trip: primal $\rightarrow$ dual $\rightarrow$ multipliers $\rightarrow$ primal point $\rightarrow$ KKT check.

Solution

Step 1. Standard form and Lagrangian. Constraint: $g(x, y) = 4 - x - y \leq 0$.

$$L(x, y, \mu) = x^2 + y^2 + \mu(4 - x - y).$$

Step 2. Inner minimization over (x, y) . L separates into a term in x and a term in y :

$$\begin{aligned} \frac{\partial L}{\partial x} = 2x - \mu = 0 &\implies x = \frac{\mu}{2}, \\ \frac{\partial L}{\partial y} = 2y - \mu = 0 &\implies y = \frac{\mu}{2}. \end{aligned}$$

Step 3. Dual function. Substitute both back into L :

$$\begin{aligned} q(\mu) &= \left(\frac{\mu}{2}\right)^2 + \left(\frac{\mu}{2}\right)^2 + \mu\left(4 - \frac{\mu}{2} - \frac{\mu}{2}\right) \\ &= \frac{\mu^2}{4} + \frac{\mu^2}{4} + \mu(4 - \mu) \\ &= \frac{\mu^2}{2} + 4\mu - \mu^2 = -\frac{\mu^2}{2} + 4\mu. \end{aligned}$$

Step 4. Solve the dual. $\max_{\mu \geq 0} -\frac{\mu^2}{2} + 4\mu$. Setting $q'(\mu) = -\mu + 4 = 0$ gives $\mu^* = 4 \geq 0$, with

$$d^* = q(4) = -\frac{16}{2} + 16 = -8 + 16 = 8.$$

Step 5. Slater's condition. The point $(3,3)$ satisfies $x + y = 6 > 4$ — strictly feasible. The objective is convex and the constraint affine, so strong duality is *guaranteed*: $p^* = d^* = 8$, before we even solve the primal.

Step 6. Recover the primal point. Evaluate the inner minimizer at $\mu^* = 4$: $x^* = y^* = 4/2 = 2$. Check: $f(2,2) = 4 + 4 = 8 = p^*$. ✓

Step 7. Verify KKT at $((2,2), \mu = 4)$.

$$\text{K1: } 2(2) - 4 = 0 \text{ (both coordinates) } \checkmark$$

$$\text{K2: } 4 - 2 - 2 = 0 \leq 0 \checkmark$$

$$\text{K3: } \mu = 4 \geq 0 \checkmark$$

$$\text{K4: } \mu \cdot g = 4 \cdot 0 = 0 \checkmark$$

The dual function $q(\mu) = -\mu^2/2 + 4\mu$: a concave hill of certified lower bounds

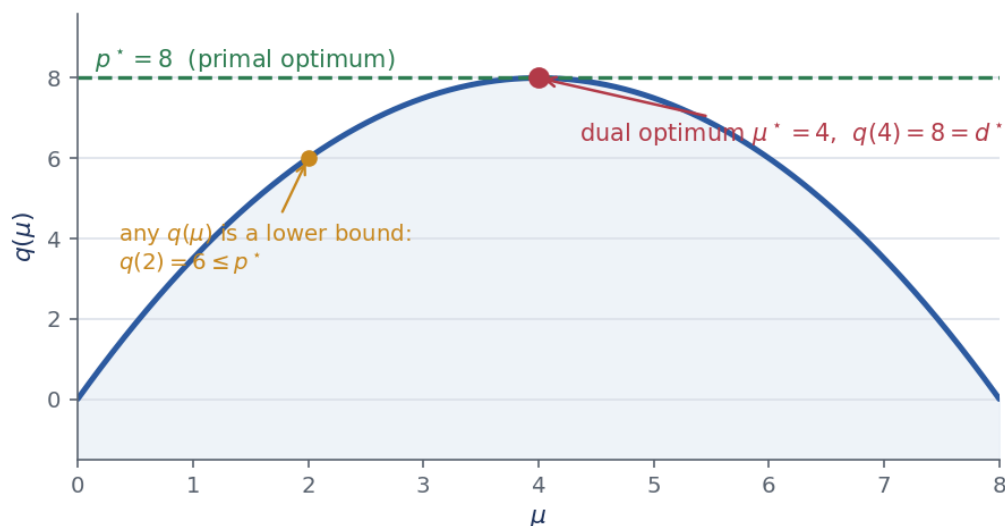


Figure 3: How to read it: the dual function $q(\mu) = -\mu^2/2 + 4\mu$ is a concave hill; every point on it is a certified lower bound on p^* (weak duality), and its peak $q(4) = 8$ exactly reaches the primal optimum (strong duality).

✓ **Final answer**

$q(\mu) = -\mu^2/2 + 4\mu$; $\mu^* = 4$, $d^* = p^* = 8$; recovered primal point $(x^*, y^*) = (2, 2)$, and all four KKT conditions hold at the pair.

* **Wait — really?**

Pick *any* $\mu \geq 0$ — say $\mu = 2$ — and compute $q(2) = -2 + 8 = 6$. Weak duality promises that 6 is a floor under the optimum of the primal, a hard guarantee obtained *without solving the primal at all*. Try any feasible point, say $(3,1)$: $f = 9 + 1 = 10 \geq 6$. Every dual value bounds every primal value. This “certificate for free” is why duality matters in practice.

Problem 13 · A dual function that is $-\infty$ for some multipliers

Consider

$$\min_x x \quad \text{subject to} \quad x^2 \leq 1.$$

Derive the dual function $q(\mu)$ — treating $\mu = 0$ and $\mu > 0$ separately — solve the dual problem, and verify strong duality.

Skill trained: handling the implicit domain of the dual: values of μ where the inner minimum runs away to $-\infty$.

Solution

Step 1. Lagrangian. $g(x) = x^2 - 1 \leq 0$ is already standard, so

$$L(x, \mu) = x + \mu(x^2 - 1) = \mu x^2 + x - \mu.$$

Step 2. Inner minimization, case $\mu = 0$. Then $L = x$, and $\inf_{x \in \mathbb{R}} x = -\infty$: with no price on the constraint, the objective just slides off to $-\infty$. So $q(0) = -\infty$ — the value $\mu = 0$ carries no information.

Step 3. Inner minimization, case $\mu > 0$. Now L is a convex parabola in x . Minimize:

$$\frac{\partial L}{\partial x} = 2\mu x + 1 = 0 \implies x = -\frac{1}{2\mu}.$$

Substitute back, term by term:

$$\begin{aligned} q(\mu) &= \mu \left(-\frac{1}{2\mu} \right)^2 + \left(-\frac{1}{2\mu} \right) - \mu \\ &= \mu \cdot \frac{1}{4\mu^2} - \frac{1}{2\mu} - \mu \\ &= \frac{1}{4\mu} - \frac{2}{4\mu} - \mu = -\frac{1}{4\mu} - \mu. \end{aligned}$$

Step 4. Solve the dual. Maximize $q(\mu) = -\frac{1}{4\mu} - \mu$ over $\mu > 0$:

$$q'(\mu) = \frac{1}{4\mu^2} - 1 = 0 \implies \mu^2 = \frac{1}{4} \implies \mu^* = \frac{1}{2} \quad (\text{taking the positive root}).$$

Then

$$d^* = q\left(\frac{1}{2}\right) = -\frac{1}{4 \cdot \frac{1}{2}} - \frac{1}{2} = -\frac{1}{2} - \frac{1}{2} = -1.$$

Step 5. Solve the primal directly. The feasible set is $x^2 \leq 1$, i.e. $-1 \leq x \leq 1$. Minimizing x over $[-1, 1]$ gives $x^* = -1$, $p^* = -1$.

Step 6. Compare and recover. $d^* = -1 = p^*$: strong duality (convex problem; $x = 0$ is strictly feasible since $0 < 1$, so Slater holds). Recovery: $x^* = -\frac{1}{2\mu^*} = -\frac{1}{2 \cdot 1/2} = -1$. ✓

✓ Final answer

$q(0) = -\infty$ and $q(\mu) = -\mu - \frac{1}{4\mu}$ for $\mu > 0$; $\mu^* = \frac{1}{2}$, $d^* = p^* = -1$, $x^* = -1$.

✗ Watch out

$q(\mu) = -\infty$ does not mean the method broke — it means that particular price fails to make the inner problem bounded, so it contributes only the useless bound $-\infty \leq p^*$. The dual problem quietly restricts itself to the multipliers where q is finite (the **domain of the dual**). Always ask: *for which μ is the inner minimum actually finite?*

✓ Key takeaway

The dual recipe never changes: (1) Lagrangian, (2) minimize over x holding μ fixed — watching for μ that send it to $-\infty$, (3) maximize $q(\mu)$ over $\mu \geq 0$, (4) recover x^* from the inner minimizer at μ^* and confirm $p^* = d^*$.

5 Part D — Judging answers, and the SVM capstone

KKT is not only a solver — it is a verifier. Feed it a candidate answer and it says yes or no, with reasons.

Problem 14 · Judging candidate answers with KKT and complementary slackness

Consider the convex problem

$$\min_{x,y} (x-3)^2 + (y-2)^2 \quad \text{subject to} \quad x \leq 2, \quad y \leq 3.$$

Three students propose primal–dual candidates $((x, y), (\mu_1, \mu_2))$:

- **Candidate A:** $(2, 2)$ with $(\mu_1, \mu_2) = (2, 0)$;
- **Candidate B:** $(2, 3)$ with $(\mu_1, \mu_2) = (2, -2)$;
- **Candidate C:** $(3, 2)$ with $(\mu_1, \mu_2) = (0, 0)$.

Test each candidate against all four KKT conditions, decide which (if any) is optimal, and state the optimal value. Then show how complementary slackness could have *predicted* the right active set without testing anything.

Skill trained: using K1–K4 as a checklist to accept or reject proposed solutions — the examiner’s view of KKT.

Solution

Step 1. Set up once. $g_1 = x - 2 \leq 0, g_2 = y - 3 \leq 0$,

$$L = (x-3)^2 + (y-2)^2 + \mu_1(x-2) + \mu_2(y-3),$$

so stationarity (K1) reads

$$2(x-3) + \mu_1 = 0 \quad \text{and} \quad 2(y-2) + \mu_2 = 0.$$

Because the problem is convex, any candidate passing K1–K4 is the global minimum; any candidate failing even one condition is out.

Step 2. Test Candidate A: $(2, 2), \mu = (2, 0)$.

$$\text{K1: } 2(2-3) + 2 = -2 + 2 = 0 \checkmark \quad 2(2-2) + 0 = 0 \checkmark$$

$$\text{K2: } g_1 = 2 - 2 = 0 \leq 0 \checkmark \quad g_2 = 2 - 3 = -1 \leq 0 \checkmark$$

$$\text{K3: } \mu_1 = 2 \geq 0 \checkmark \quad \mu_2 = 0 \geq 0 \checkmark$$

$$\text{K4: } \mu_1 g_1 = 2 \cdot 0 = 0 \checkmark \quad \mu_2 g_2 = 0 \cdot (-1) = 0 \checkmark$$

All four hold. **A is optimal.**

Step 3. Test Candidate B: $(2, 3), \mu = (2, -2)$. Stationarity actually holds: $2(2-3) + 2 = 0$ and $2(3-2) + (-2) = 0$. Primal feasibility holds ($g_1 = 0, g_2 = 0$), and so does K4 (both constraints tight). But K3 fails:

$$\mu_2 = -2 < 0. \quad \mathbf{X}$$

Reject B — dual infeasibility. Reading: to hold y up at the boundary $y = 3$, the constraint would have to *pull the point away from* where the objective wants to go

($y = 2$ is already feasible) — a wall can only push, never pull, and $\mu_2 < 0$ is the algebra saying exactly that.

Step 4. Test Candidate C: $(3, 2)$, $\mu = (0, 0)$. Stationarity holds: $2(3 - 3) + 0 = 0$ and $2(2 - 2) + 0 = 0$; K3 and K4 hold trivially. But K2 fails:

$$g_1 = 3 - 2 = 1 \not\leq 0. \quad \times$$

Reject C — primal infeasibility. C is just the unconstrained minimizer with the constraints ignored.

Step 5. Optimal value. From A: $f(2, 2) = (2 - 3)^2 + (2 - 2)^2 = 1 + 0 = 1$.

Step 6. The complementary-slackness shortcut. The unconstrained minimizer is $(3, 2)$. It violates only g_1 (x -limit), and satisfies g_2 strictly. So predict: g_1 active, g_2 inactive — i.e. $x = 2$, $\mu_2 = 0$. Then K1 instantly gives $y = 2$ (from $2(y - 2) + 0 = 0$) and $\mu_1 = -2(x - 3) = -2(2 - 3) = 2 \geq 0$. That is Candidate A, derived in two lines rather than discovered by testing.

✓ **Final answer**

Only Candidate A passes all four KKT conditions: $(x^*, y^*) = (2, 2)$, $(\mu_1^*, \mu_2^*) = (2, 0)$, $f^* = 1$. B fails dual feasibility ($\mu_2 < 0$); C fails primal feasibility ($x > 2$).

✗ **Watch out**

A candidate can pass *three* conditions and still be wrong — B passed stationarity, primal feasibility *and* complementary slackness. Never declare optimality until all four boxes are ticked; in an exam, write the four checks explicitly, one line each.

📖 **Problem 15 · Capstone: a tiny hard-margin SVM, from primal to support vectors**

A one-dimensional dataset has two points: $x_1 = +1$ with label $y_1 = +1$, and $x_2 = -1$ with label $y_2 = -1$. The hard-margin SVM problem is

$$\min_{w, b} \frac{1}{2} w^2 \quad \text{subject to} \quad y_i (wx_i + b) \geq 1, \quad i = 1, 2.$$

- Write both constraints explicitly and form the Lagrangian with multipliers $\alpha_1, \alpha_2 \geq 0$.
- Derive the dual problem and solve it.
- Recover w^* and b^* , identify the support vectors using complementary slackness, and give the decision boundary and margin.

Skill trained: every skill in this set, chained: frame $\rightarrow$ standard form $\rightarrow$ Lagrangian $\rightarrow$ dual $\rightarrow$ solve $\rightarrow$ recover $\rightarrow$ read complementary slackness.

Solution

Step 1. Write the constraints. For $i = 1$: $y_1(wx_1 + b) = (+1)(w \cdot 1 + b) = w + b \geq 1$. For $i = 2$: $y_2(wx_2 + b) = (-1)(-w + b) = w - b \geq 1$. In standard form:

$$g_1 = 1 - w - b \leq 0, \quad g_2 = 1 - w + b \leq 0.$$

Step 2. Lagrangian.

$$L(w, b, \alpha_1, \alpha_2) = \frac{1}{2}w^2 + \alpha_1(1 - w - b) + \alpha_2(1 - w + b).$$

Step 3. Stationarity in w and b .

$$\begin{aligned}\frac{\partial L}{\partial w} &= w - \alpha_1 - \alpha_2 = 0 \implies w = \alpha_1 + \alpha_2, \\ \frac{\partial L}{\partial b} &= -\alpha_1 + \alpha_2 = 0 \implies \alpha_1 = \alpha_2.\end{aligned}$$

(These are the general SVM identities $w = \sum_i \alpha_i y_i x_i$ and $\sum_i \alpha_i y_i = 0$, written out for this data.)

Step 4. Build the dual function. Group L as

$$L = \frac{1}{2}w^2 - w(\alpha_1 + \alpha_2) - b(\alpha_1 - \alpha_2) + (\alpha_1 + \alpha_2).$$

Set $\alpha_1 = \alpha_2 = \alpha$ (so the b -term vanishes) and substitute $w = \alpha_1 + \alpha_2 = 2\alpha$:

$$\begin{aligned}q(\alpha) &= \frac{1}{2}(2\alpha)^2 - (2\alpha)(2\alpha) + 2\alpha \\ &= 2\alpha^2 - 4\alpha^2 + 2\alpha = 2\alpha - 2\alpha^2.\end{aligned}$$

(Check against the textbook dual $\sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j x_i x_j$: here every product $y_i y_j x_i x_j$ equals 1, so it reads $(\alpha_1 + \alpha_2) - \frac{1}{2}(\alpha_1 + \alpha_2)^2 = 2\alpha - 2\alpha^2$. Same function.)

Step 5. Solve the dual. Maximize $q(\alpha) = 2\alpha - 2\alpha^2$ over $\alpha \geq 0$:

$$q'(\alpha) = 2 - 4\alpha = 0 \implies \alpha^* = \frac{1}{2} \geq 0, \quad d^* = q\left(\frac{1}{2}\right) = 1 - \frac{1}{2} = \frac{1}{2}.$$

$$\text{So } \alpha_1^* = \alpha_2^* = \frac{1}{2}.$$

Step 6. Recover w^* and b^* . $w^* = \alpha_1^* + \alpha_2^* = \frac{1}{2} + \frac{1}{2} = 1$. Since $\alpha_1^* = \frac{1}{2} > 0$, complementary slackness forces its constraint tight: $w + b = 1$, so $b^* = 1 - w^* = 0$. Consistency: the second constraint gives $w - b = 1 - 0 = 1$ — also tight, matching $\alpha_2^* > 0$. Primal value: $\frac{1}{2}(w^*)^2 = \frac{1}{2} = d^*$: zero duality gap. ✓**Step 7. Read the geometry.** Decision boundary: $w^*x + b^* = 0$, i.e. $x = 0$ — exactly midway between the two points. Margin lines: $w^*x + b^* = \pm 1$, i.e. $x = \pm 1$ — passing through the data points. Margin width $= 2/|w^*| = 2$. Both points sit on the margin with $\alpha_i > 0$: by complementary slackness, both are **support vectors**.**✓ Final answer**

Dual: $\max_{\alpha \geq 0} 2\alpha - 2\alpha^2$ with $\alpha_1 = \alpha_2 = \alpha$; solution $\alpha_1^* = \alpha_2^* = \frac{1}{2}$, $d^* = \frac{1}{2}$. Recovered classifier: $w^* = 1$, $b^* = 0$, decision boundary $x = 0$, margin width 2. Both data points are support vectors.

✓ Key takeaway

Complementary slackness is the reason SVMs are “sparse”: a training point with $\alpha_i = 0$ sits strictly outside the margin and could be deleted without changing the classifier; only the points with $\alpha_i > 0$ — the support vectors, whose constraints are tight — carry the

The tiny SVM: $w^* = 1$, $b^* = 0$ — both points sit exactly on the margin

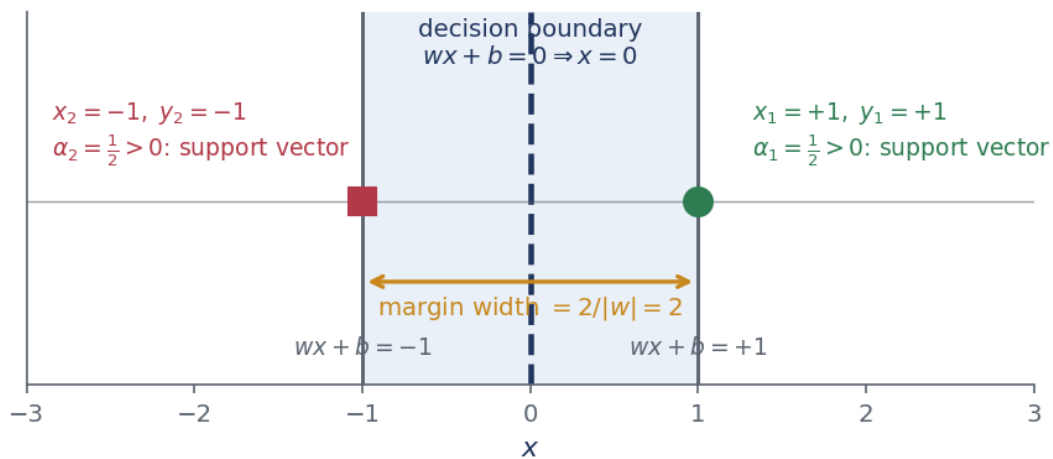


Figure 4: How to read it: the classifier is the midpoint $x = 0$; the shaded band is the margin, whose edges $x = \pm 1$ pass exactly through the two data points — that is what $\alpha_i > 0 \Rightarrow$ constraint tight looks like in a picture.

solution. The multiplier tells you which data matter.

6 Close the loop: five-question self-check

Answer from memory first; the answers are printed upside down.

Q1. At the optimum you find $\mu_3 = 0$. What does complementary slackness let you conclude about $g_3(x^*)$?

Nothing beyond feasibility. $\mu_3 g_3 = 0$ is satisfied by $\mu_3 = 0$ whatever $g_3 \leq 0$ is — the constraint may be active or inactive. It is only $\mu_i > 0$ that forces $g_i = 0$ (and $g_i > 0$ that forces $\mu_i = 0$).

Q2. True or false: if the primal problem is nonconvex, the dual function $q(\mu)$ may fail to be concave.

False. For each fixed x , $L(x, \mu)$ is affine in μ ; the infimum of affine functions is always concave. The dual is a concave maximization even when the primal is ugly — that is much of its charm.

Q3. Prove weak duality in one line, for feasible x and $\mu \geq 0$.

$q(\mu) = \inf_z L(z, \mu) \leq L(x, \mu) = f(x) + \mu g(x) \leq f(x)$, because $\mu \geq 0$ and $g(x) \leq 0$ make $\mu g(x) \leq 0$. Take the best on both sides: $p^* \leq p$.

Q4. In an SVM, what does $\alpha_i = 0$ mean geometrically, and what may you do with that training point?

The point lies strictly outside the margin ($y_i(w x_i + b) > 1$). Its constraint is slack, so it exerts no force on the solution — delete it and the classifier does not move. It is not a support vector.

Q5. Your case analysis (Recipe 3) returns $\mu = -3$ for a constraint you assumed active. What do you do?

Discard the whole case — dual feasibility (K3) fails. The assumed active set is wrong: the constraint is trying to “pull” rather than push. Move to the next active/inactive pattern.

7 Pocket card: the whole topic on one page

If you remember only this page, you can rebuild everything else.

Move	What you actually do
Standard form	$\min f$; every inequality as $g_i \leq 0$; every equality as $h_j = 0$. $\max f \rightarrow \min(-f)$; $a \geq b \rightarrow b - a \leq 0$.
Lagrangian	$L = f + \sum_i \mu_i g_i + \sum_j \lambda_j h_j$ ($\mu_i \geq 0$; λ_j free).
Equality-only shortcut	$\nabla f = \lambda \nabla h$ together with $h = 0$ (tangency of level set and constraint).
KKT checklist	K1 $\nabla_x L = 0$; K2 $g_i \leq 0$, $h_j = 0$; K3 $\mu_i \geq 0$; K4 $\mu_i g_i = 0 \forall i$. Convex problem: K1–K4 $\Rightarrow$ global optimum.
Case method	Each g_i is active ($g_i = 0$) or inactive ($\mu_i = 0$). Solve each pattern; discard on <i>infeasibility</i> , <i>negative multiplier</i> , or a <i>contradiction inside K1</i> ; keep what survives.
Dual function	$q(\mu, \lambda) = \inf_x L(x, \mu, \lambda)$ — an <i>unconstrained</i> minimization; note where $q = -\infty$. Always concave.
Dual problem	$\max q$ over $\mu \geq 0$ (and free λ). Weak duality: $d^* \leq p^*$, always. Strong duality: convex + Slater (a strictly feasible point) $\Rightarrow d^* = p^*$.
Recovery	Plug μ^* into the inner minimizer to get x^* ; confirm $f(x^*) = q(\mu^*)$ and run the KKT checklist on the pair.
Meanings	λ = shadow price (df^*/db); μ = force of a wall (zero if untouched); $\alpha_i > 0$ = support vector.

✓ Key takeaway

Frame it, standardize it, write the Lagrangian, and then let the four KKT lines — or the dual — do the thinking. Every one of the fifteen problems above is these same moves in a different costume.